

Creating "The Great Question of Life" Game in Python

In this guide, you will learn how to create a simple text-based game in Python inspired by "The Hitchhiker's Guide to the Galaxy". The game asks the player to guess the answer to the ultimate question of life, the universe, and everything.

Creating the Main Game Script

Let's start by creating the main game script step by step. We will explain each part of the code in detail as we build the function.

Checking the User's Answer

Next, let's write a function to check the player's answer and respond accordingly. We'll start with defining the correct answers.

```
1 correct_answers = ["42", "forty-two", "forty two"]
```

Listing 1: Defining correct answers

Explanation:

- `correctanswers = ["42", "forty-two", "forty two"]`: This list contains all acceptable correct answers.

Prompting for User Input

First, we prompt the player to input their answer.

```
1 answer = input("What is your answer? ")
```

Listing 2: Prompting for user input

Explanation:

- `answer = input("What is your answer? ")`: This line prompts the player to input their answer and stores it in the variable `answer`.

Formatting the Answer

Next, we format the answer to ensure the comparison is case-insensitive and ignores leading and trailing spaces.

```
1 formatted_answer = answer.strip().lower()
```

Listing 3: Formatting the answer

Explanation:

- `.strip()`: This ignores leading and trailing spaces by stripping the answer of any white spaces.
- `.lower()`: This ensures that the answer comparison is case-insensitive, by making all characters into lowercase.

Checking if the Answer is Correct

We then check if the formatted answer is in the list of correct answers.

```
1 if formatted_answer in correct_answers:
```

Listing 4: Checking if the answer is correct

Explanation:

- `if formatted_answer in correct_answers`: This line checks if the formatted answer is in the list of correct answers.

Handling a Correct Answer

If the answer is correct, we display a correct answer message and print a confirmation message.

```
1 print("Yes, 42 is indeed the answer to the ultimate question of  
    life, the universe, and everything!")
```

Listing 5: Handling a correct answer

Explanation:

- `print("Yes, 42 is indeed the answer to the ultimate question of life, the universe, and everything!")`: This line prints a confirmation message if the answer is correct.

Handling an Incorrect Answer

If the answer is incorrect, we display an incorrect answer message and print a message indicating the answer is incorrect.

```
1 print("Incorrect answer human")
```

Listing 6: Handling an incorrect answer

Explanation:

- `print("Incorrect answer human")`: This line prints a message indicating the answer is incorrect.

Now let's encapsulate this logic in a function called `check_answer`.

```
1 def check_answer(answer):
2     correct_answers = ["42", "forty-two", "forty two"]
3     if answer.strip().lower() in correct_answers:
4         return "Yes, 42 is indeed the answer to the ultimate
5         question of life, the universe, and everything!"
6     else:
7         return "Incorrect answer human"
```

Listing 7: Encapsulating logic in a function

Explanation:

- `def check_answer(answer)`: This line defines the function `check_answer`, which takes one parameter, `answer`.
- The rest of the code inside the function is the same logic we discussed earlier.

Creating the Main Function

Now, let's write the main function to run the game.

```
1 def main():
2     print("In 'The Hitchhiker's Guide to the Galaxy' by Douglas
3     Adams,")
4     print("a supercomputer named Deep Thought determines the answer
5     to the ultimate question of life, the universe, and everything
6     .")
7     user_answer = input("Can you guess what is the answer to the
8     Great Question of Life, the Universe, and Everything? ")
9     print(check_answer(user_answer))
```

Listing 8: Defining the main function

Explanation:

- `print()`: These lines display the game's introductory text.
- `user_answer = input()`: This line asks the player to input their answer.
- `print(check_answer(user_answer))`: This line checks the player's answer by calling our function `check_answer` and prints the result it returns.

Running the Game

Finally, we need to run the main function when the script is executed.

```
1 if __name__ == "__main__":
2     main()
```

Listing 9: Running the main function

Explanation:

- `if __name__ == "__main__":`: This line checks if the script is being run directly (not imported as a module).
- `main()`: This runs the main function.

Your `main.py` file should now look like this:

```
1
2 def check_answer(answer):
3     correct_answers = ["42", "forty-two", "forty two"]
4     if answer.strip().lower() in correct_answers:
5         return "Yes, 42 is indeed the answer to the ultimate
6             question of life, the universe, and everything!"
7     else:
8         return "Incorrect answer human"
9
10 def main():
11     print("In 'The Hitchhiker's Guide to the Galaxy' by Douglas
12         Adams,")
13     print("a supercomputer named Deep Thought determines the answer
14         to the ultimate question of life, the universe, and everything
15         .")
16     user_answer = input("Can you guess what is the answer to the
17         Great Question of Life, the Universe, and Everything? ")
18     print(check_answer(user_answer))
19
20 if __name__ == "__main__":
21     main()
```

Listing 10: Complete main.py file

Explanation:

- `if __name__ == "__main__":`: This line checks if the script is being run directly (not imported as a module).
- `main()`: This runs the main function.

Adding Extra Flavor

To make the game more engaging, we'll add visual representations of the correct and incorrect answers using ASCII art. We'll store these in a separate file called `render.py`.

1. Create a new file named `render.py`.
2. Add the following functions to display the correct and incorrect messages.

```
def render_correct():
    print("""
        ~+

                *      +
            ,      |
        ( )  .-.,=" ' "=-.  - o -
            '=/_      \      |
        *    | '=._      |
            \      '=. / ' ,
            . '=._.= ' '= ' *
        +      0      *      ,      +
    """)

def render_incorrect():
    print("""
        -----
        _dMMMb._      .ad0000000000ba.      _dMMMb._
        dP' ~YMMb      d000000000000000b      aMMP~ 'Yb
        V ~"Mb      d00000000000000000b      dM"~ V
            'Mb.      d00000000000000000b      ,dM'
            'YMb._ |0000000000000000000| _dMP'
            'YMMM| OP'~"Y0000000000OP"~'YO |MMMP'
        __      'YMMM| OP'~"Y0000000000OP"~'YO |MMMP'
        ,dMMMb. ~~~' 00 'Y00000P' 00 '~~ ,dMMMb.
        _dP~ 'YMba_ 00b '000' d00 _aMMP' ~Yb._
            'YMMM\ '000o 000 o000' /MMMP'
        ,aa. 'YMMb '000b._,d000b._,d000'dMMP~ ,aa.
        ,dMYMb._ '0000000000000000000' _adMYMb.
        ,MP' 'YMMba._ 000000000000000000 _adMMP' 'YM.
        MP' ~YMMMba._ Y0000PVVVVVY0000P _adMMMP~ 'YM
        YMb ~YMMM\ '0000I' ' ' ' ' I00000' /MMMP~ dMP
            'Mb. 'YMMb' 000I , , , , I0000'dMMP' ,dM'
            ' , '00bNNNNNd00' ,
            '~00000~'
    """)
```

To use the functions we created in `render.py`, we need to import the file into our main file:

```
1 import render # This will allow us to use the functions we created
```

Listing 11: Importing necessary modules

Now we need to specify where and when the `render_correct` and `render_incorrect` functions from `render.py` will be used. We will do that by calling each one when the player's answer is either correct or incorrect, respectively.

Now your `main.py` file should look like this:

```
1 import render # This will allow us to use the functions we created
2
3 def check_answer(answer):
4     correct_answers = ["42", "forty-two", "forty two"]
5     if answer.strip().lower() in correct_answers:
6         render.render_correct() # This will call the render_correct
           function
7         return "Yes, 42 is indeed the answer to the ultimate
           question of life, the universe, and everything!"
8     else:
9         render.render_incorrect() # This will call the
           render_incorrect function
10        return "Incorrect answer human"
11
12 def main():
13     print("In 'The Hitchhiker's Guide to the Galaxy' by Douglas
           Adams,")
14     print("a supercomputer named Deep Thought determines the answer
           to the ultimate question of life, the universe, and everything
           .")
15     user_answer = input("Can you guess what is the answer to the
           Great Question of Life, the Universe, and Everything? ")
16     print(check_answer(user_answer))
17
18 if __name__ == "__main__":
19     main()
```

Listing 12: Complete main.py file with render.py module integration

That's it! You now have a fully functional text-based game that asks the ultimate question of life, the universe, and everything.